EXPERIMENT 5

Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

1. Aim

To analyze the S-parameter response of a microwave directional coupler and a two-way power divider over 8–12 GHz, and to evaluate coupling, isolation, insertion loss, directivity and equal power division at a design frequency of 10 GHz.

2. Objectives

1. To design microwave directional-coupler and power-divider structures for a specified operating frequency.
2. To determine coupling factor, isolation, insertion loss and directivity of the directional coupler.
3. To analyze power division ratio, port matching and output-port isolation of the power divider.
4. To optimize the physical dimensions for improved S-parameter performance over the desired operating bandwidth.

3. Parameters Used

Parameter	Value
Design frequency	10 GHz
Reference impedance	50 Ω
Frequency sweep	8–12 GHz
Number of frequency points	401
Coupling target	10 dB
Isolation target	≥ 25 dB
Insertion-loss target	≤ 1.5 dB
Directivity target	≥ 15 dB
Length change for comparison	5%
Ideal divider output levels	−3.01 dB each

